Smart Shopping Cart System

Advanced Support Squad



September 2025

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1. SYSTEM OVERVIEW

1.1 Objective

1.1.1 Problem Statement



Figure 1.1: Customers unable to find desired products at the mall.



Người dân Hà Nội đi mua sắm tại hệ thống siêu thị AEON Mall Hà Đông chiều tối 3/2. Ảnh: Ngọc Thành

Figure 1.2: Crowds pack a Hanoi supermarket ahead of Tet.

- Hard to find products
- Lack of product information
- Staff may not always be available

- No tech support
- Long lines at checkout counters
- Issues with checking bills

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1. SYSTEM OVERVIEW

1.1 Objective

1.1.1 Solution Overview

- Indoor navigation support
- Product recognition
- User-friendly interface
- Voice command
- Recognizing customer behavior
- …

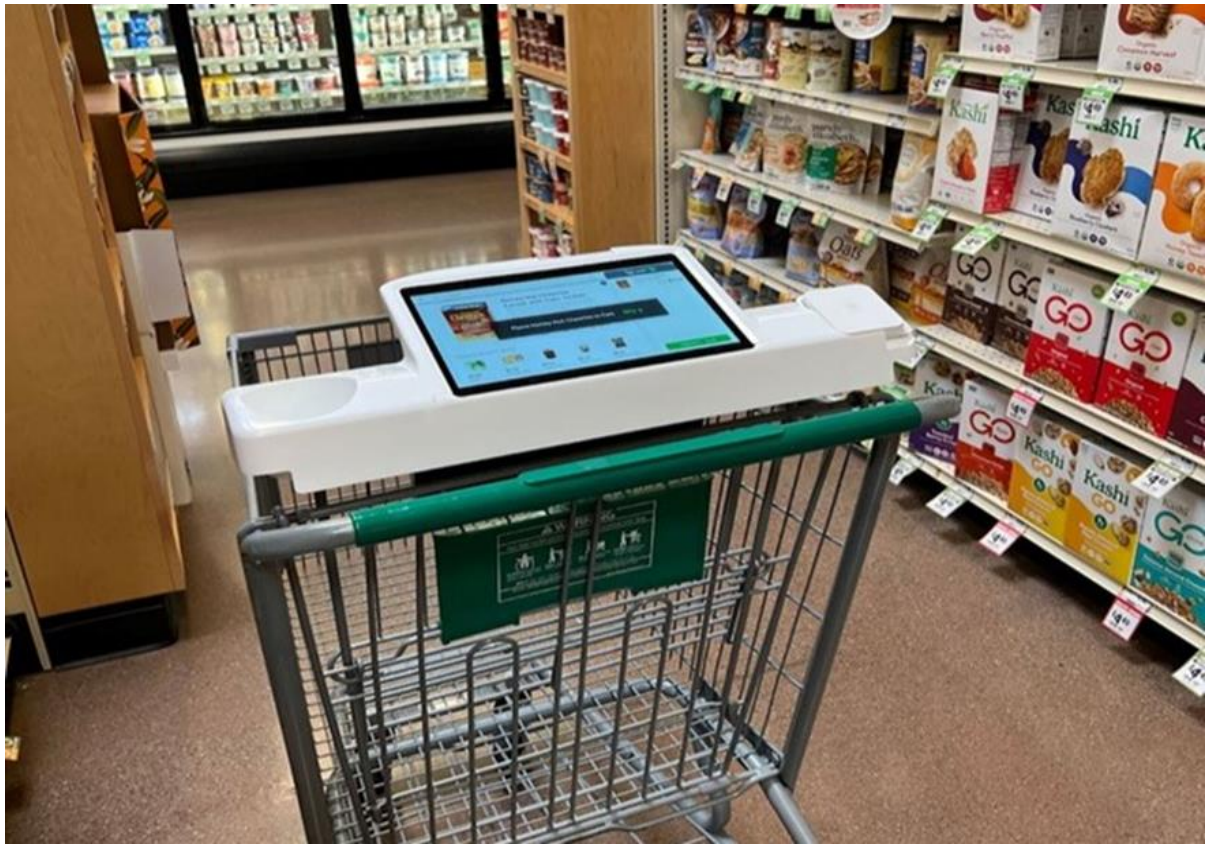


Figure 1.3: Smart shopping cart system

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1. SYSTEM OVERVIEW

1.2 End users and Clients

- **End users**: Customers shopping in supermarkets and retail stores.
- **Business clients**: Large supermarkets, retail chains, and commercial centers.



Figure 1.4: Image of the smart shopping cart system in real-world use.



Figure 1.5: Japan Self-checkout Suppermarket cart system.

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1.3 System Diagram

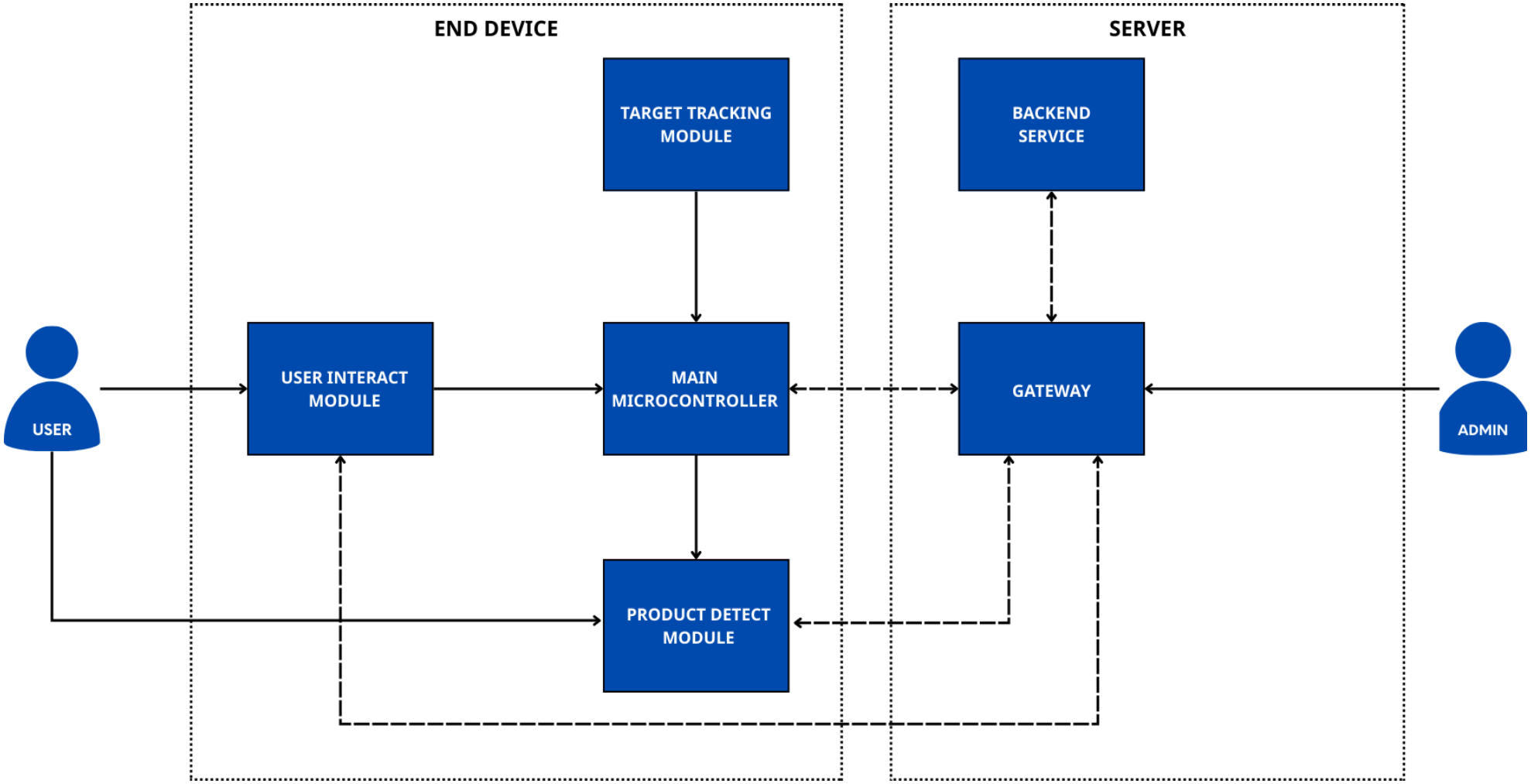


Figure 1.6: System module diagram

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1. SYSTEM OVERVIEW

1.3 System Diagram

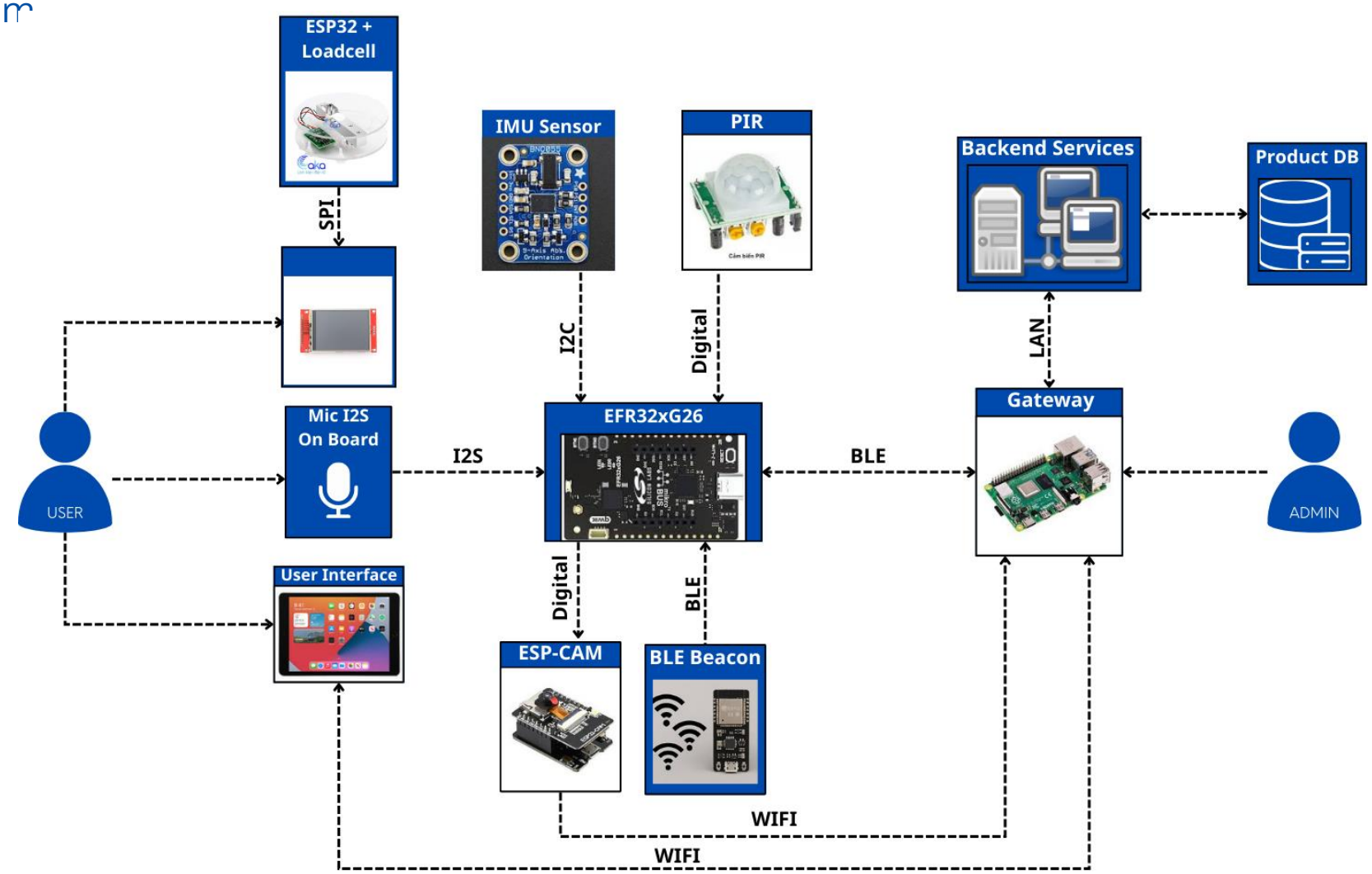


Figure 1.7: System device and comunication diagram

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2. MAIN FEATURES

2.1 Indoor Positioning

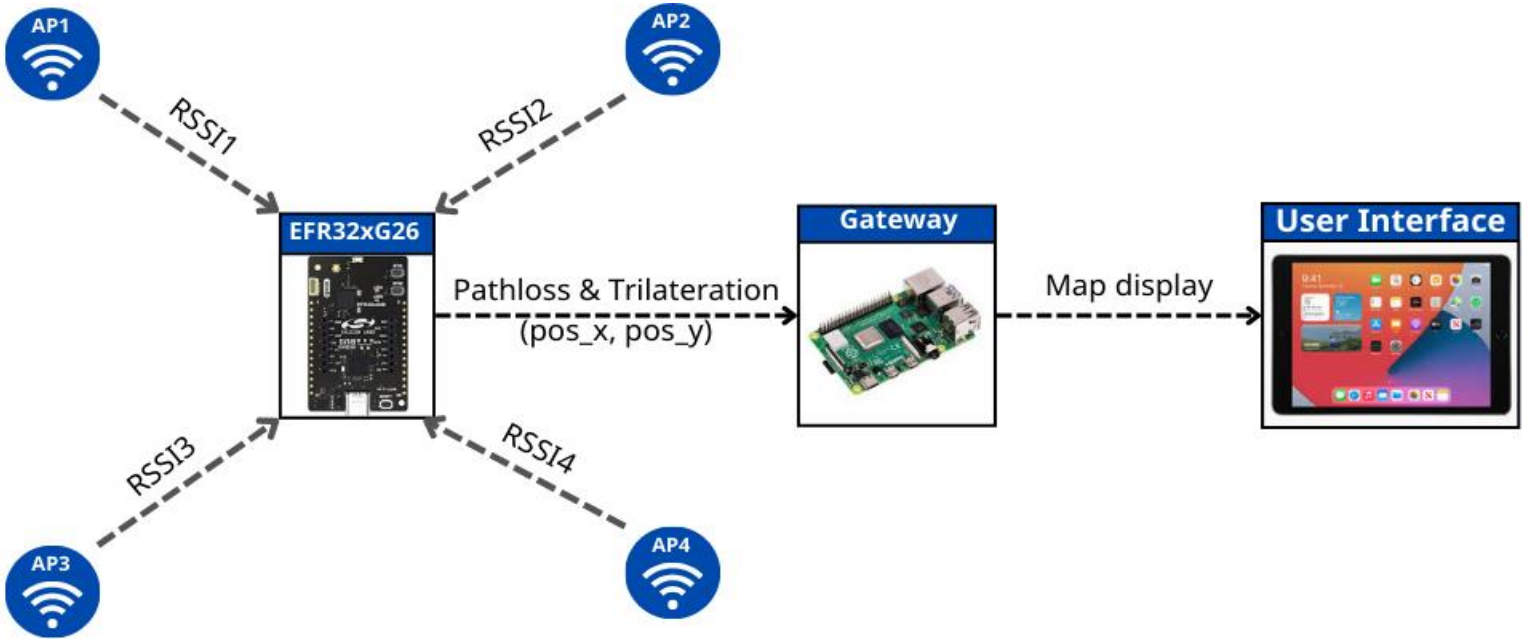


Figure 2.1: In"mall" Positioning System.

In-Mall Navigation:

- **Objective:** Helps users easily track their real-time location, find products, and navigate to new areas on the map
- **Implementation:** The EFR32 collects RSSI values from surrounding beacons, then calculates and updates the shopping cart's position on the map in real time.

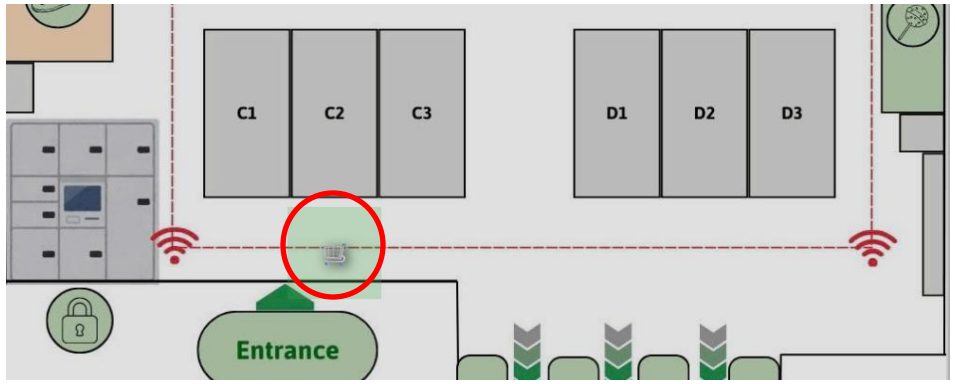


Figure 2.2: User IPS interface.

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2. MAIN FEATURES

2.1 Indoor Positioning

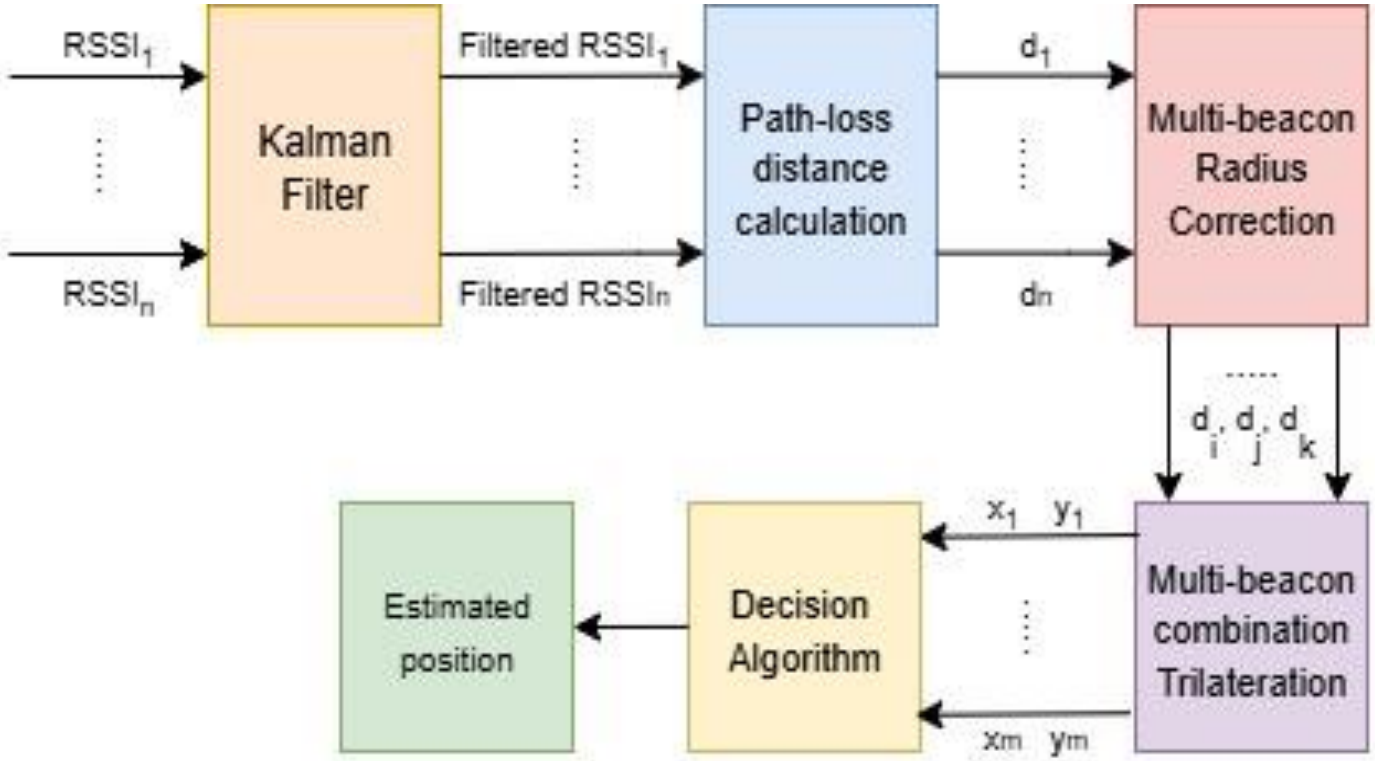


Figure 2.3: IPS Algorithm

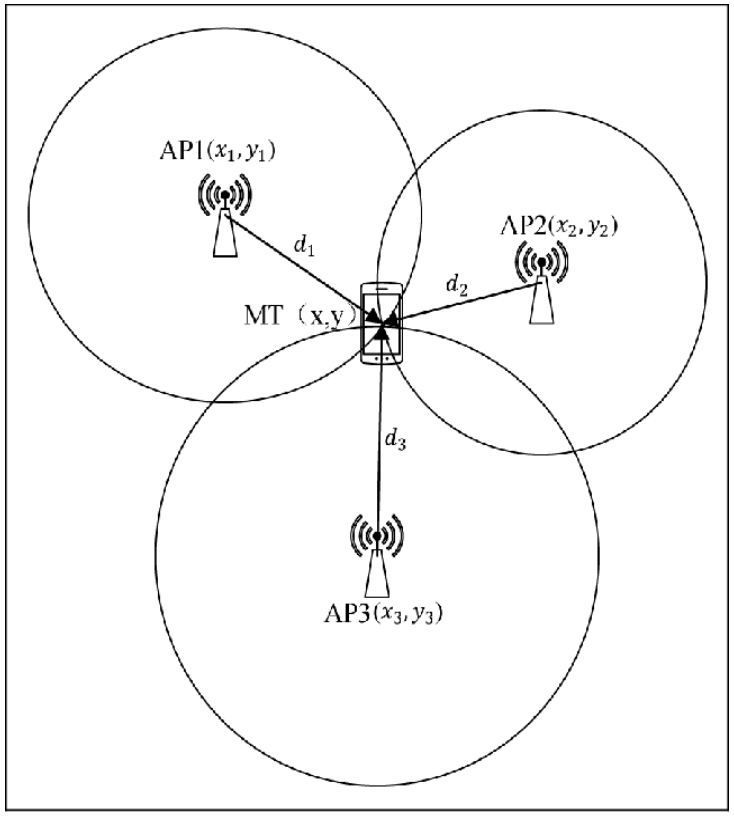


Figure 2.4: Trilateration Algorithm

In real-world deployment, due to environmental noise and the limited number of only four beacons, the algorithm achieved a positioning accuracy ranging from 0.5 to 2 meters — a significantly higher precision compared to conventional GPS systems.

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2. MAIN FEATURES

2.2 Voice Command – Edge AI

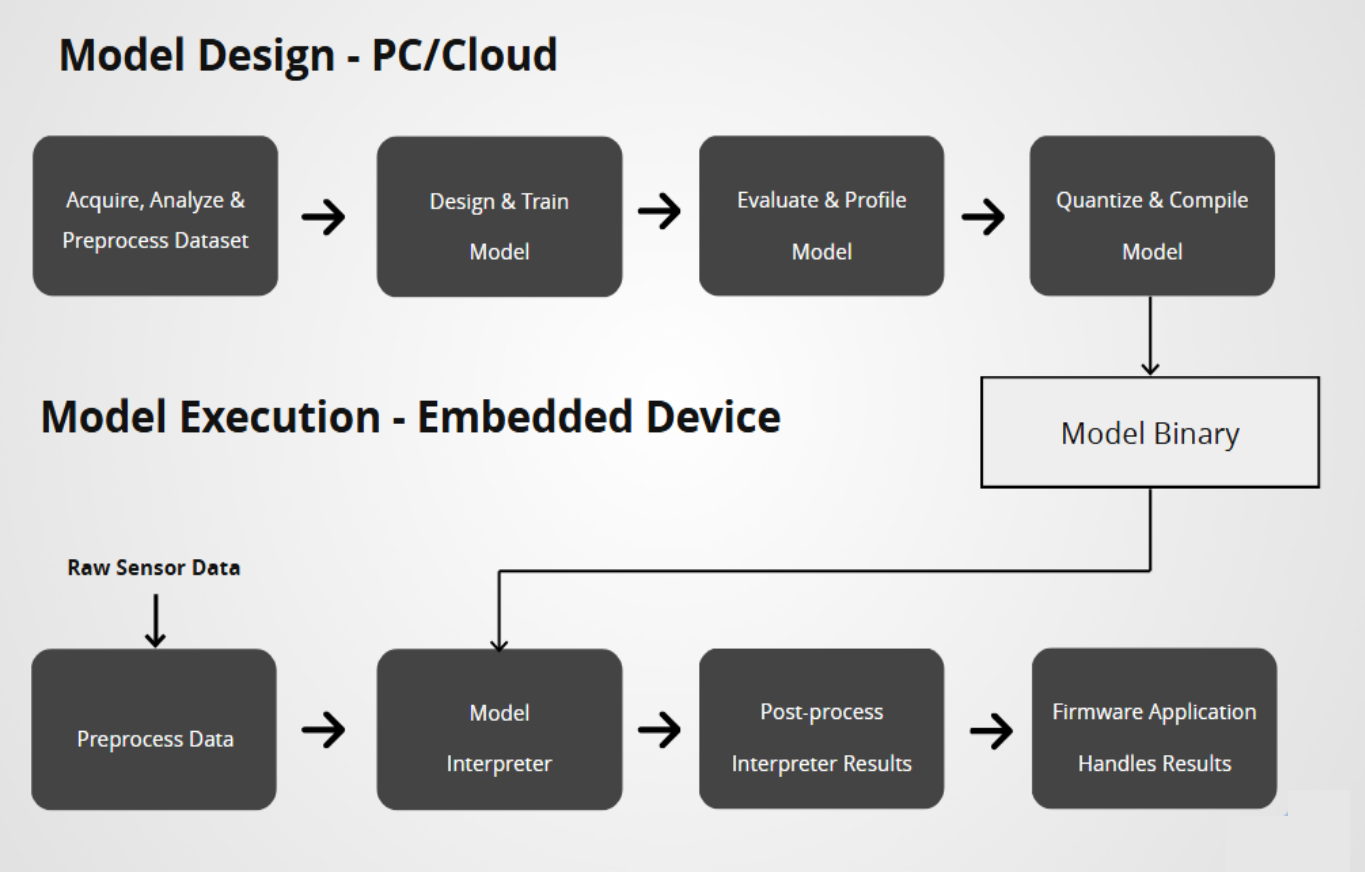


Figure 2.5: Edge AI trainning model

- **Objective:**
  - Allows users to search for areas on the map using their voice.
  - No staff assistance

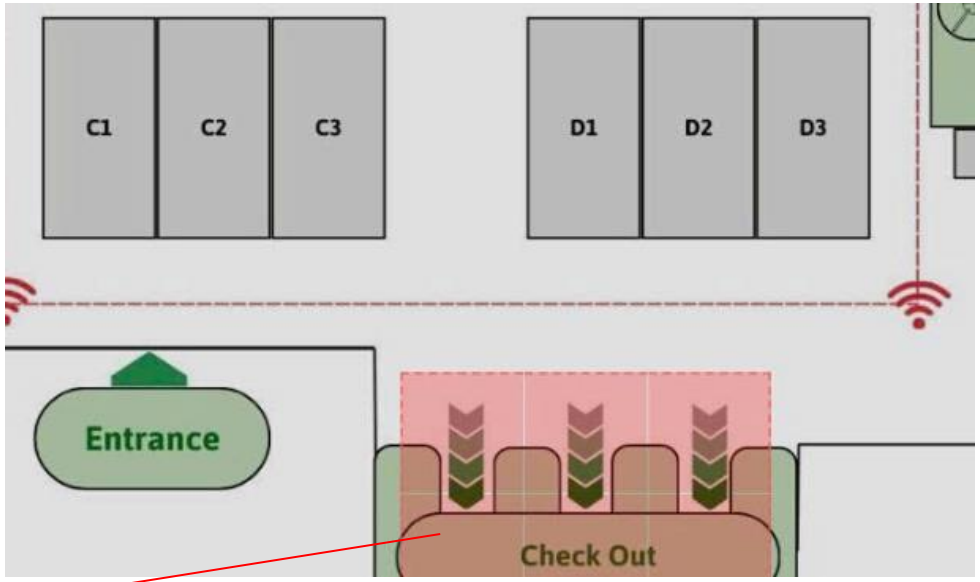


Figure 2.5: Highlight map when user use voice command

Highlight when the voice command "checkout" is detected.

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2. MAIN FEATURES

2.3 Behavior Recognition

- **Goal:** Enhance safety and usage monitoring

- **Sensors used:**
  - **PIR Sensor** → Detects hand movement into the cart (adding/removing items)
  - **IMU BNO055** → Detects cart motion and unusual activities

- **Behaviors be recognized:**
  - Hand in the cart → CAM on
  - Cart tilting or falling → Warning!
  - Strong shaking or vibration → Warning!

- The recognized actions can be observed through the LEDs on the circuit.
  - **CAM on** → Green led on
  - **Warning** → Red led on



Figure 2.6: Behavior Recognition module

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2. MAIN FEATURES

2.4 Products Detection

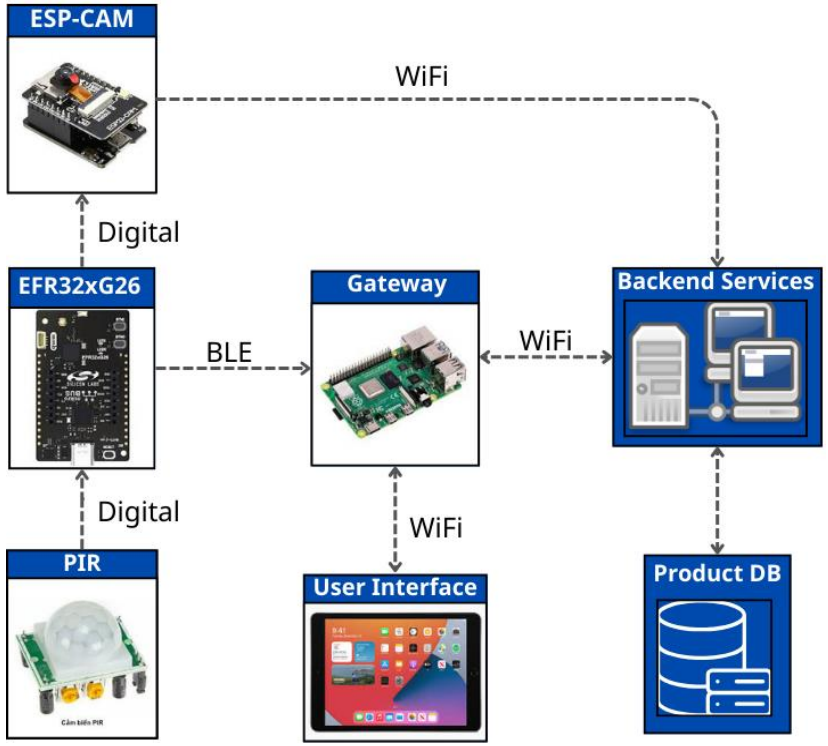


Figure 2.6:Products detect structure

Products Live Searching:

- Instantly identify products by scanning product packaging with a live camera.
- Provides an easy way for users to check and manage their items without manual input.

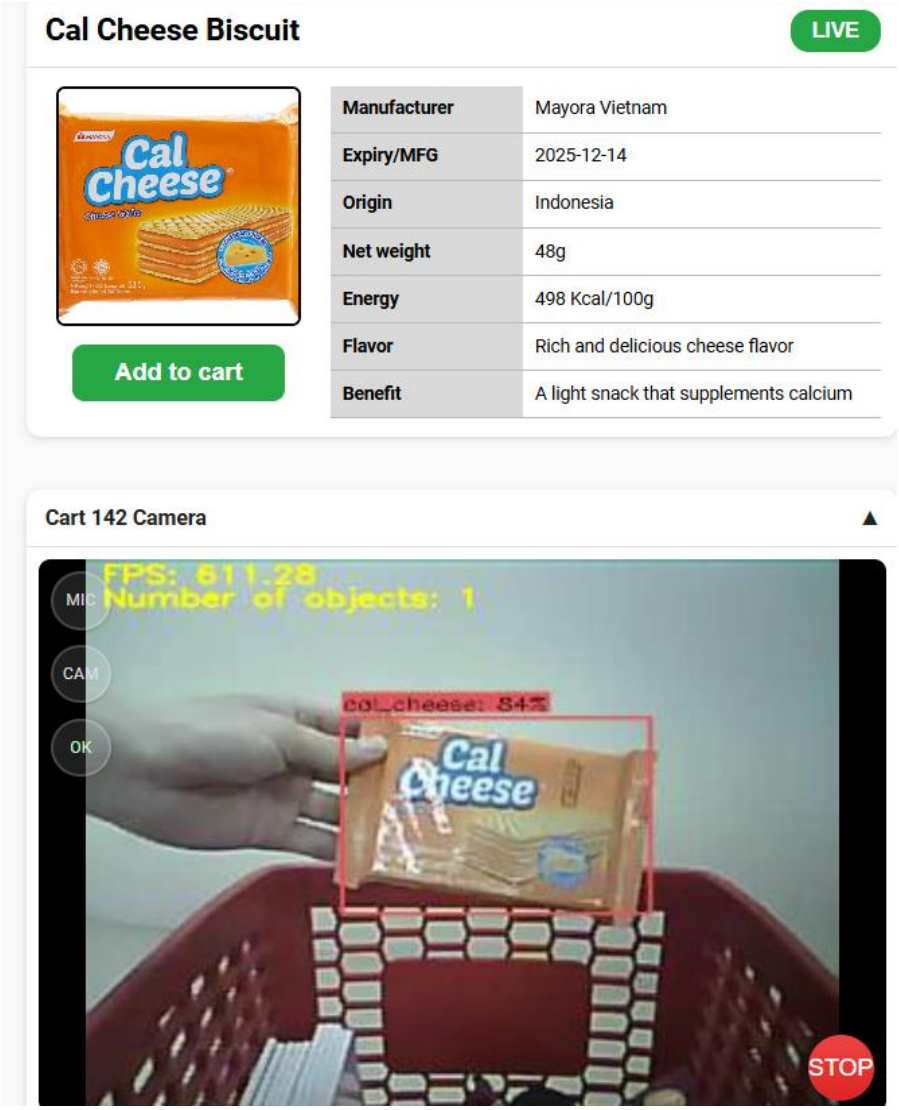


Figure 2.7:Products detect user interface

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3. PRODUCT IMPLEMENTATION

3.1.1 Cart-User Interface



Figure 3.1: Cart Hardware Interface

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3. PRODUCT IMPLEMENTATION

3.1.2 Shopping list-User Interface

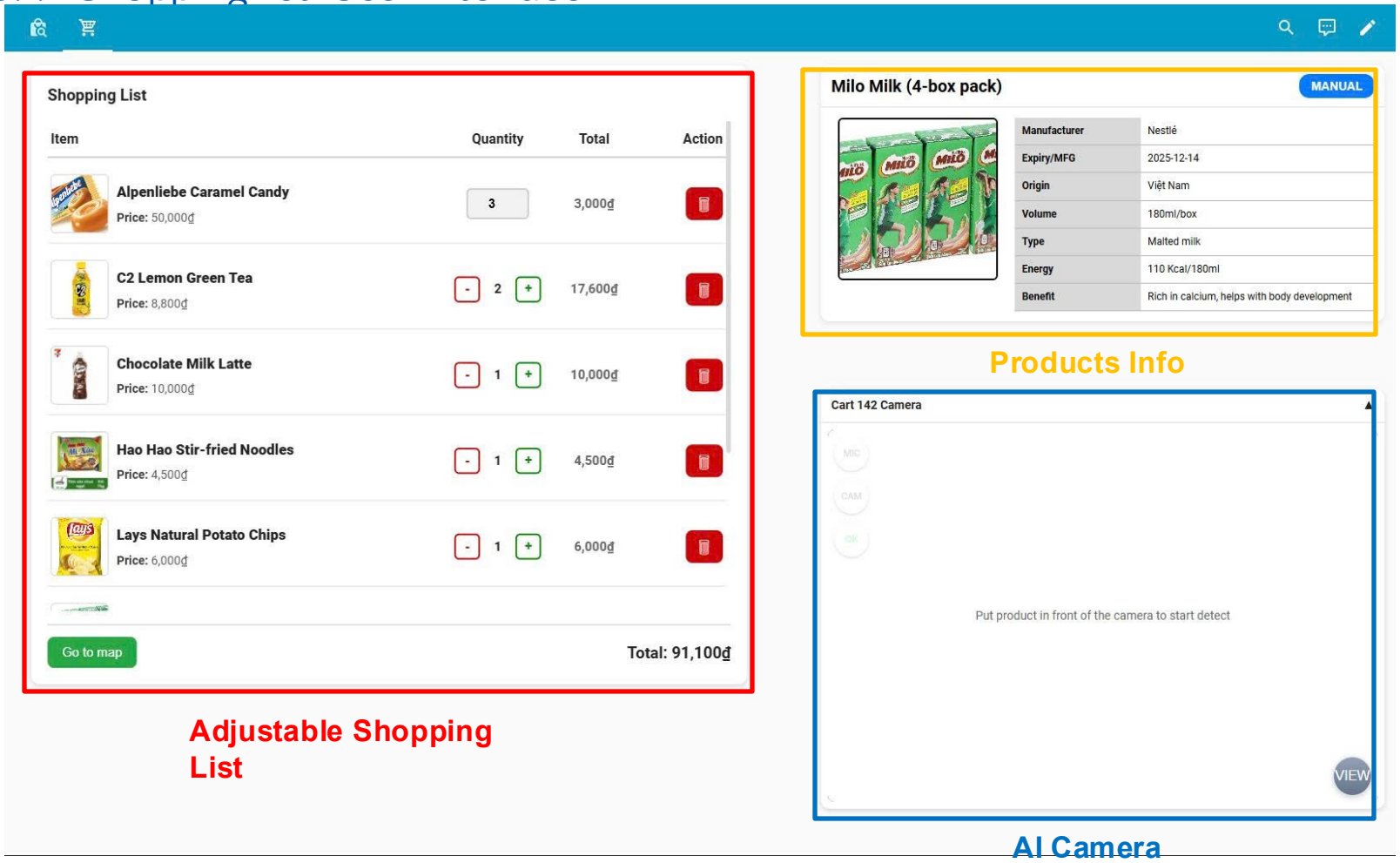


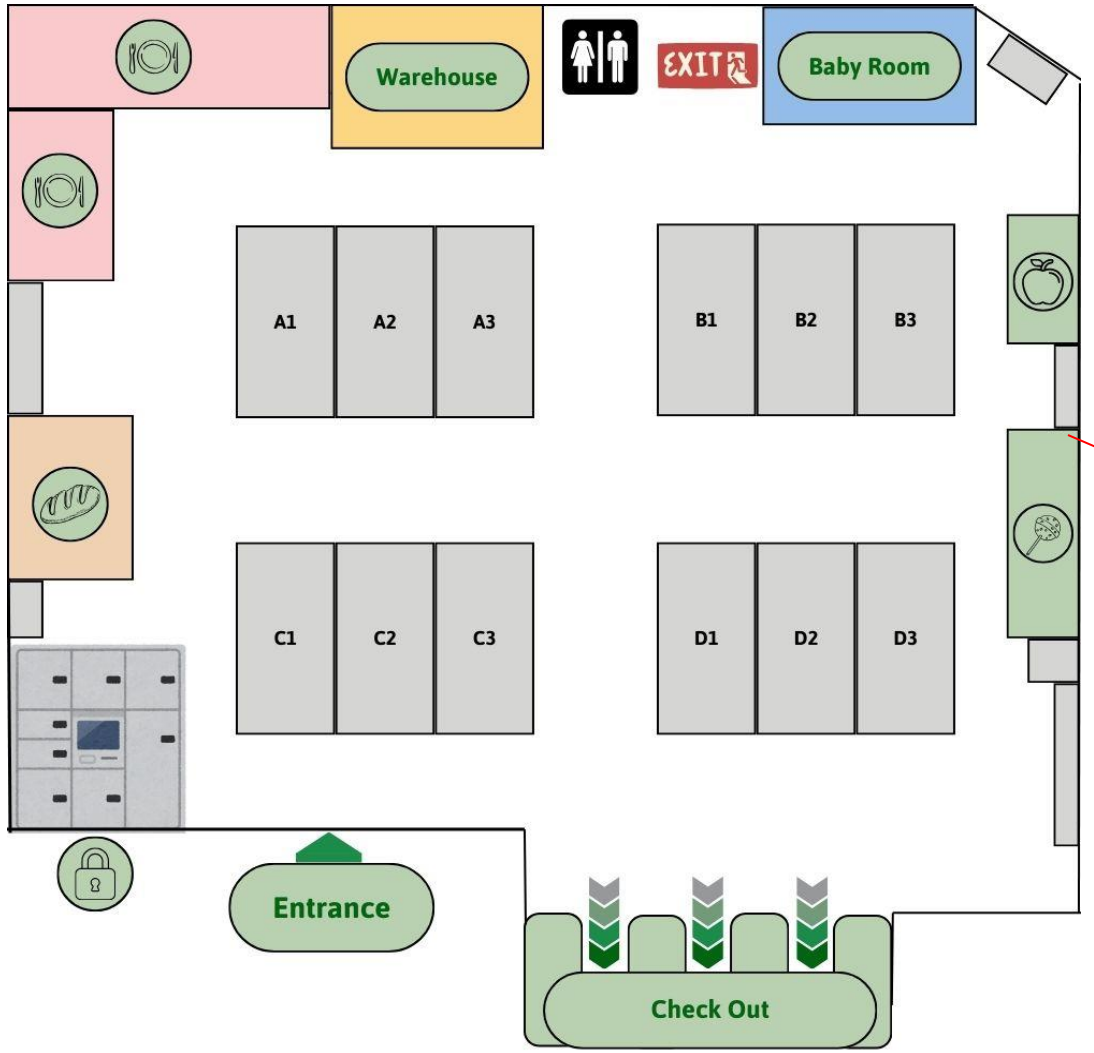
Figure 3.2: Shopping list features

- **Adjustable Shopping List:** Adjust items and quantities and view real-time subtotals and totals. Tap any item to see its details.
- **Product Info:** Displays product details from two sources: Tapping on an item in the list or scanning a product with the AI camera
- **AI camera:** Shows a real-time view from the in-cart camera, allowing users to scan products for AI detection

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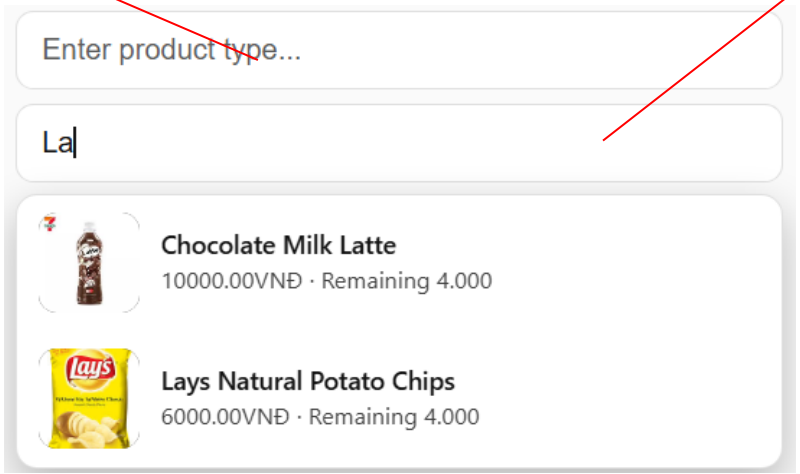
3. PRODUCT IMPLEMENTATION

3.1.3 Map & Search – User Interface

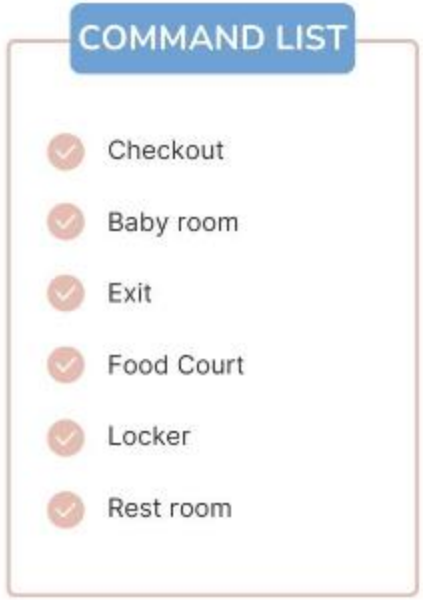


Search by type

Search by name



Pop-up



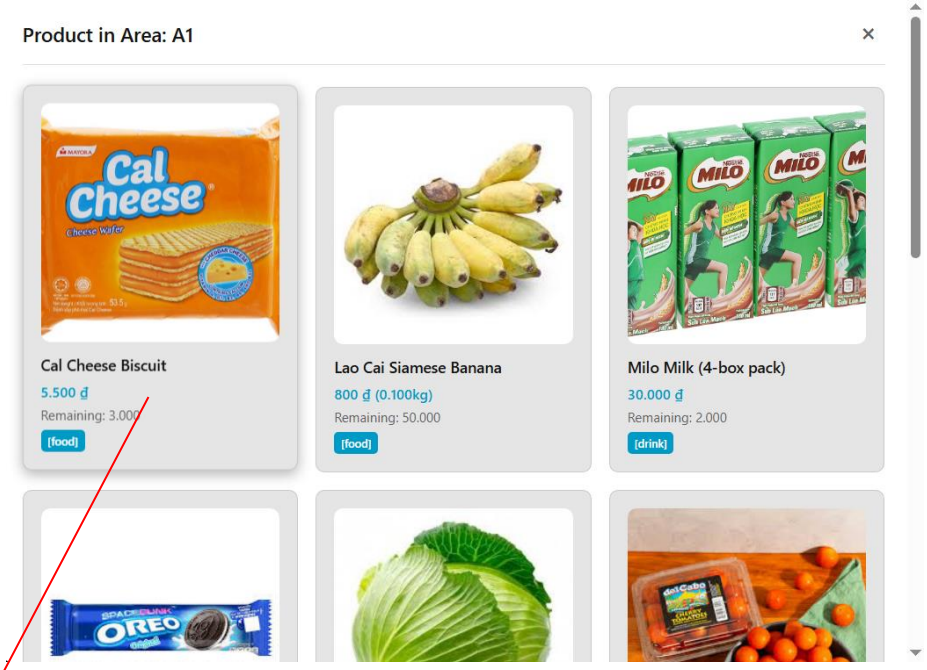
- **Other features**
  - Show the user's location in map
  - Highlight areas with voice commands

Figure 3.3: Map layout & Search features

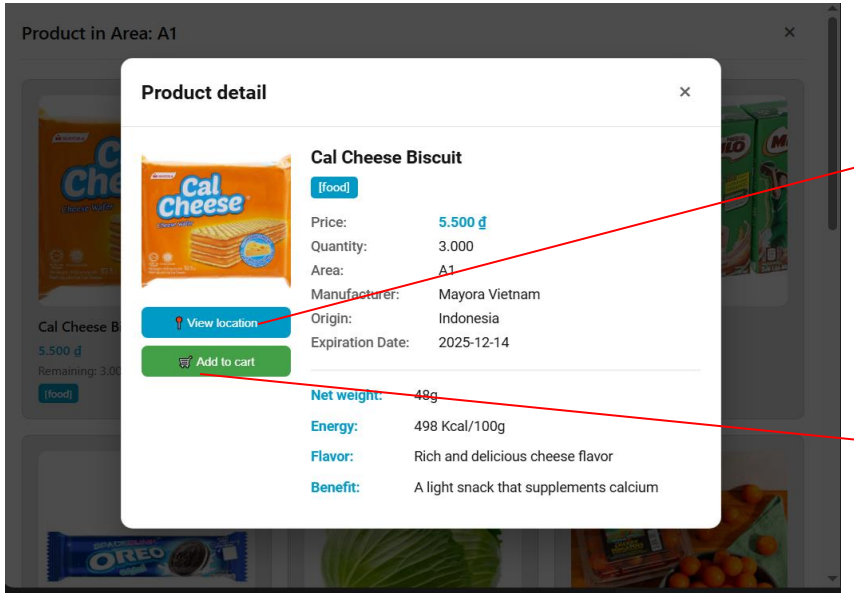
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3. PRODUCT IMPLEMENTATION

3.1.3 Map & Search – User Interface

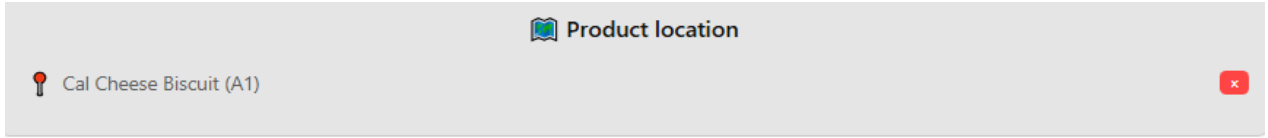


1. Select a product



2. View location

3. Add to cart





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3. PRODUCT IMPLEMENTATION

3.2.1 Product manage – Admin Interface

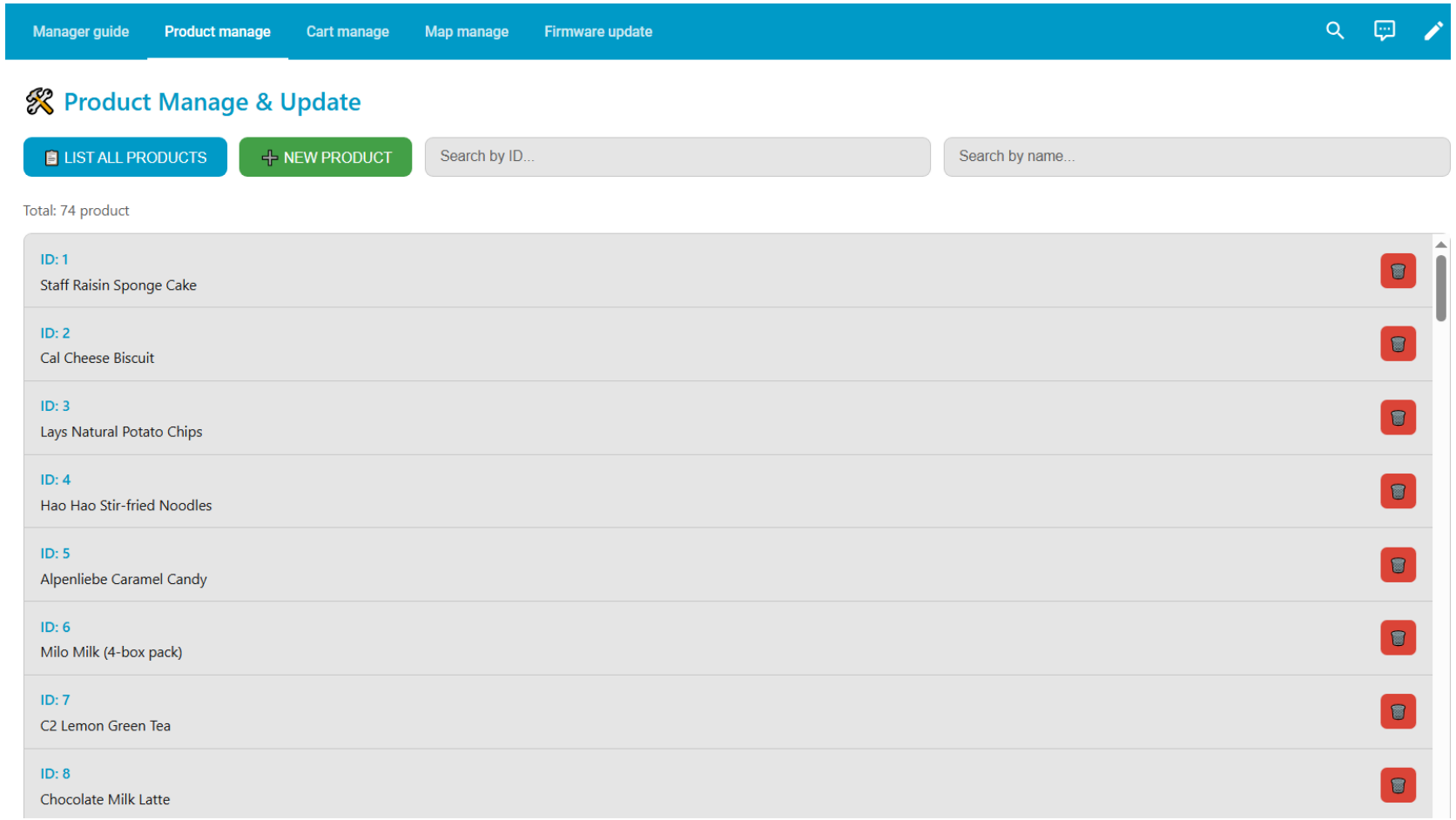


Figure 3.4: Product manage page

Features:

- View the total number of products in the warehouse.
- Display a list of all products with ID and name.
- Add or remove products to the warehouse.
- Search for products by ID or by name
- Update and manage existing product information.

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3. PRODUCT IMPLEMENTATION

3.2.1 Product manage – Admin Interface

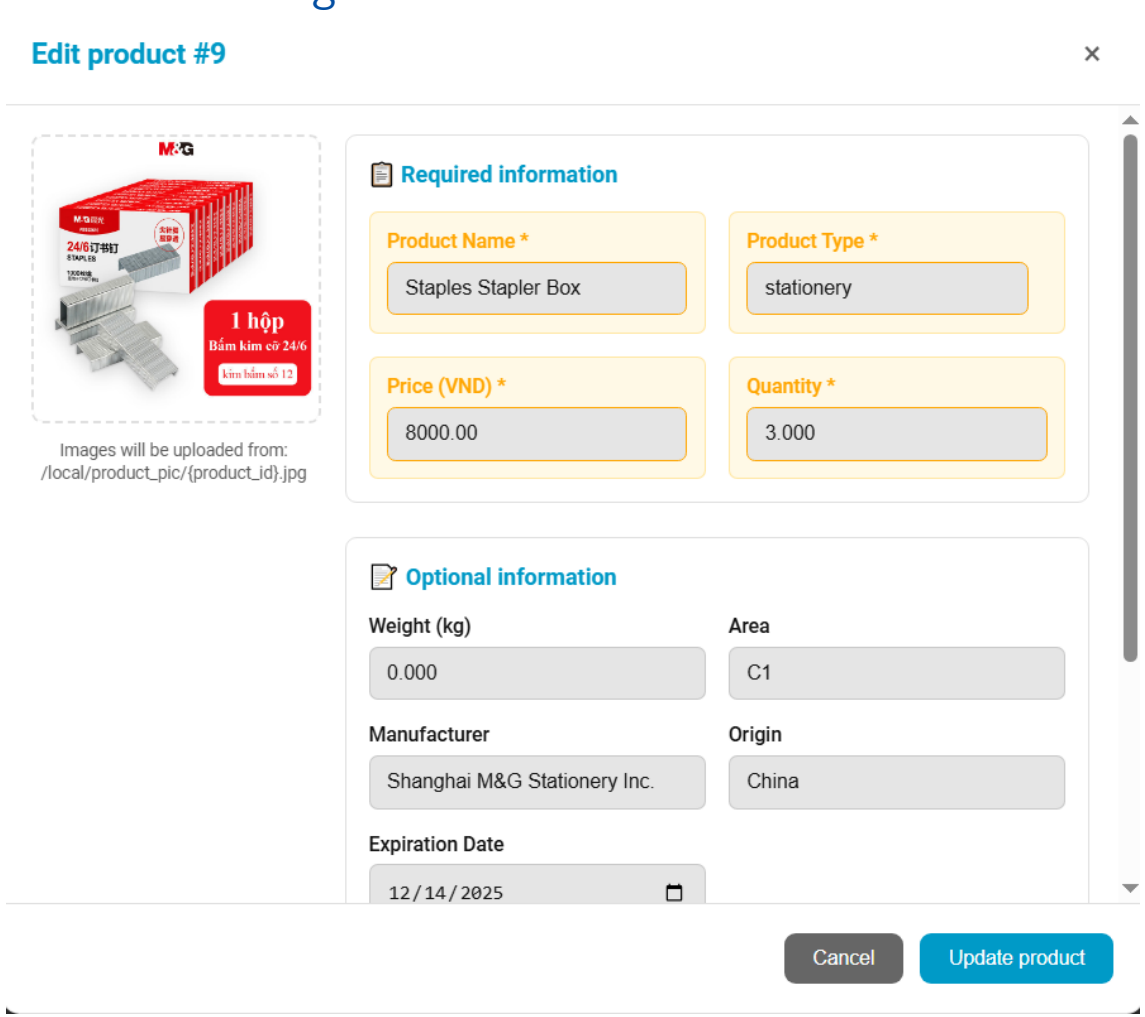


Figure 3.5: Update products information

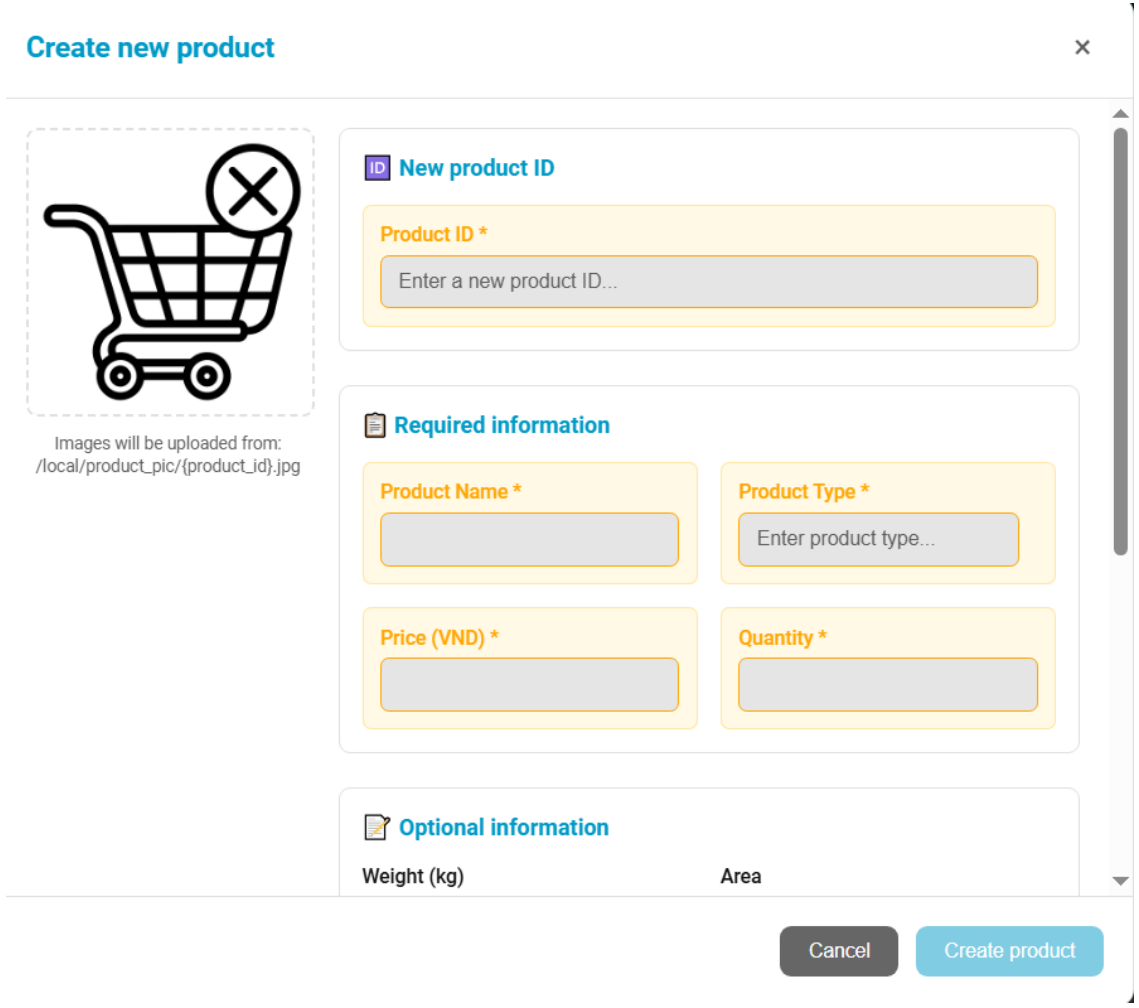


Figure 3.6: Create a new product

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3. PRODUCT IMPLEMENTATION

3.2.2 Cart manage-Admin Interface

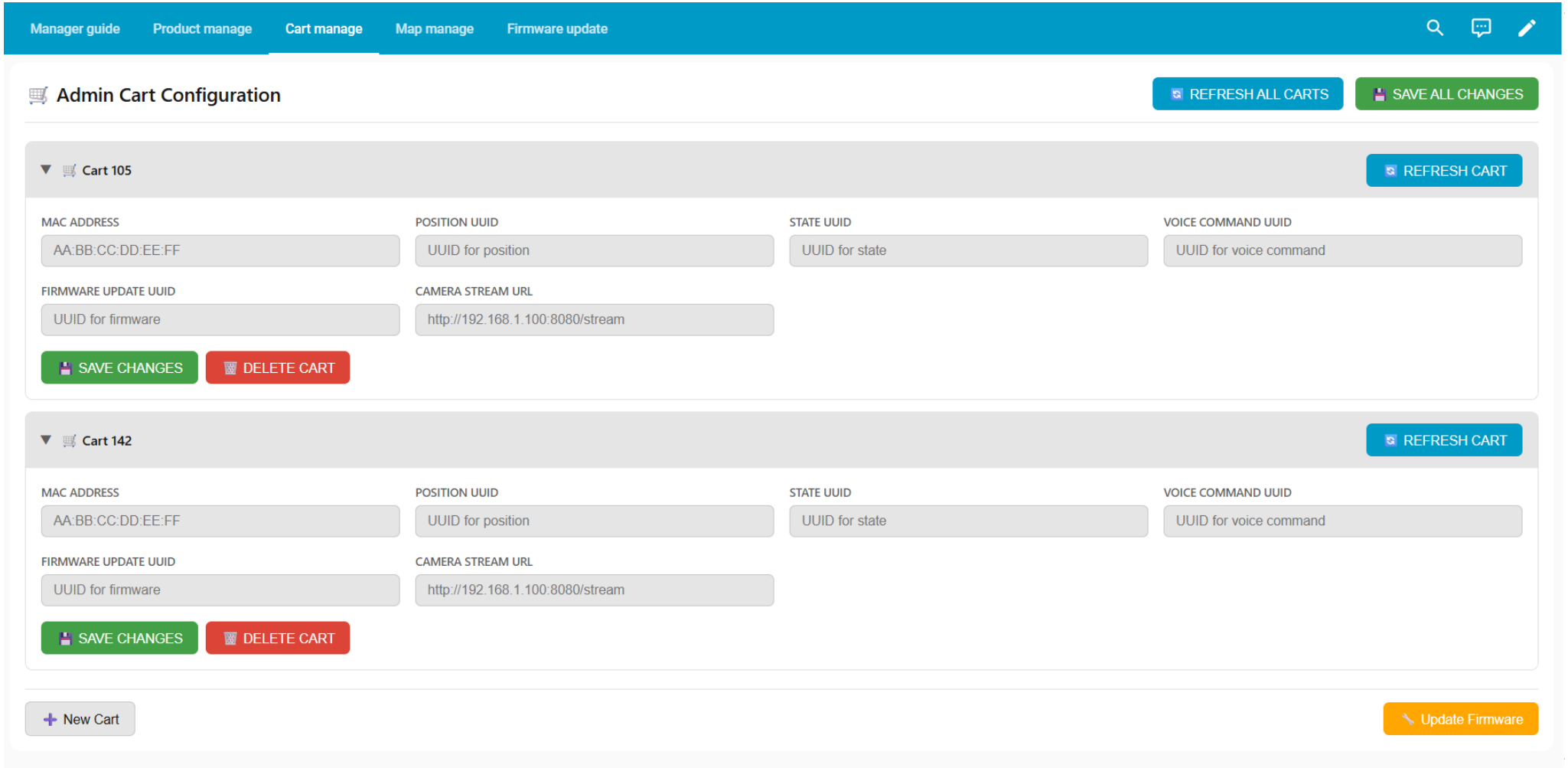


Figure 3.7: Cart manage page interface.

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3. PRODUCT IMPLEMENTATION

3.2.3 Map manage-Admin Interface

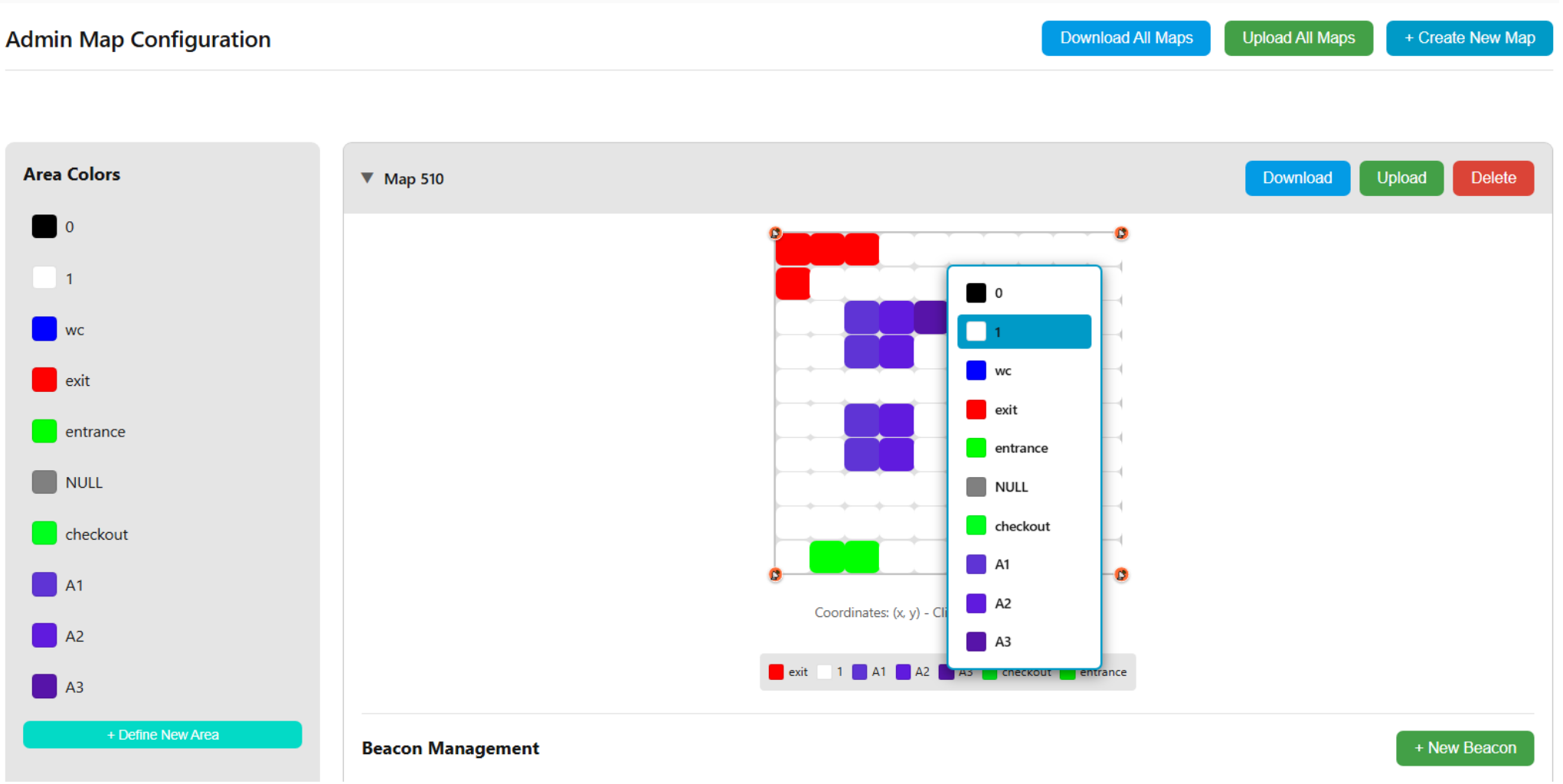


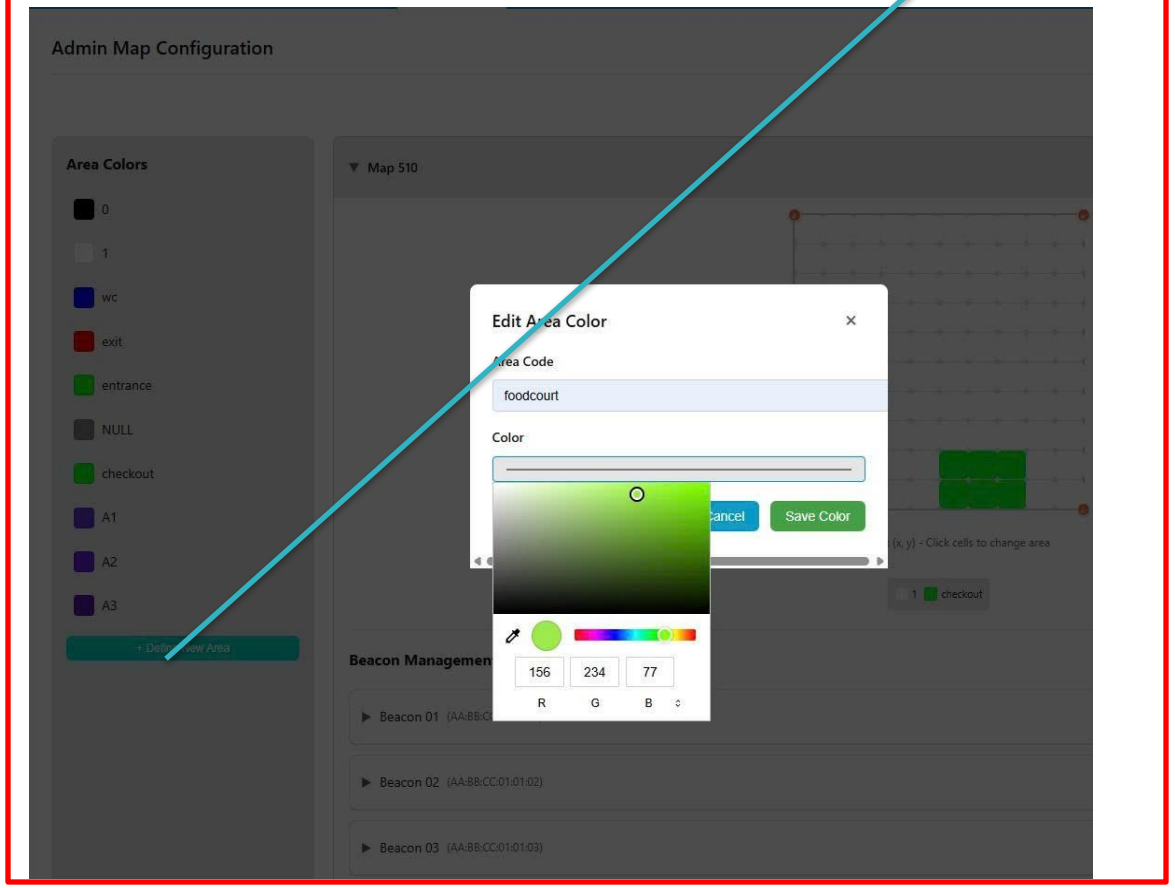
Figure 3.8: Map manage page interface.

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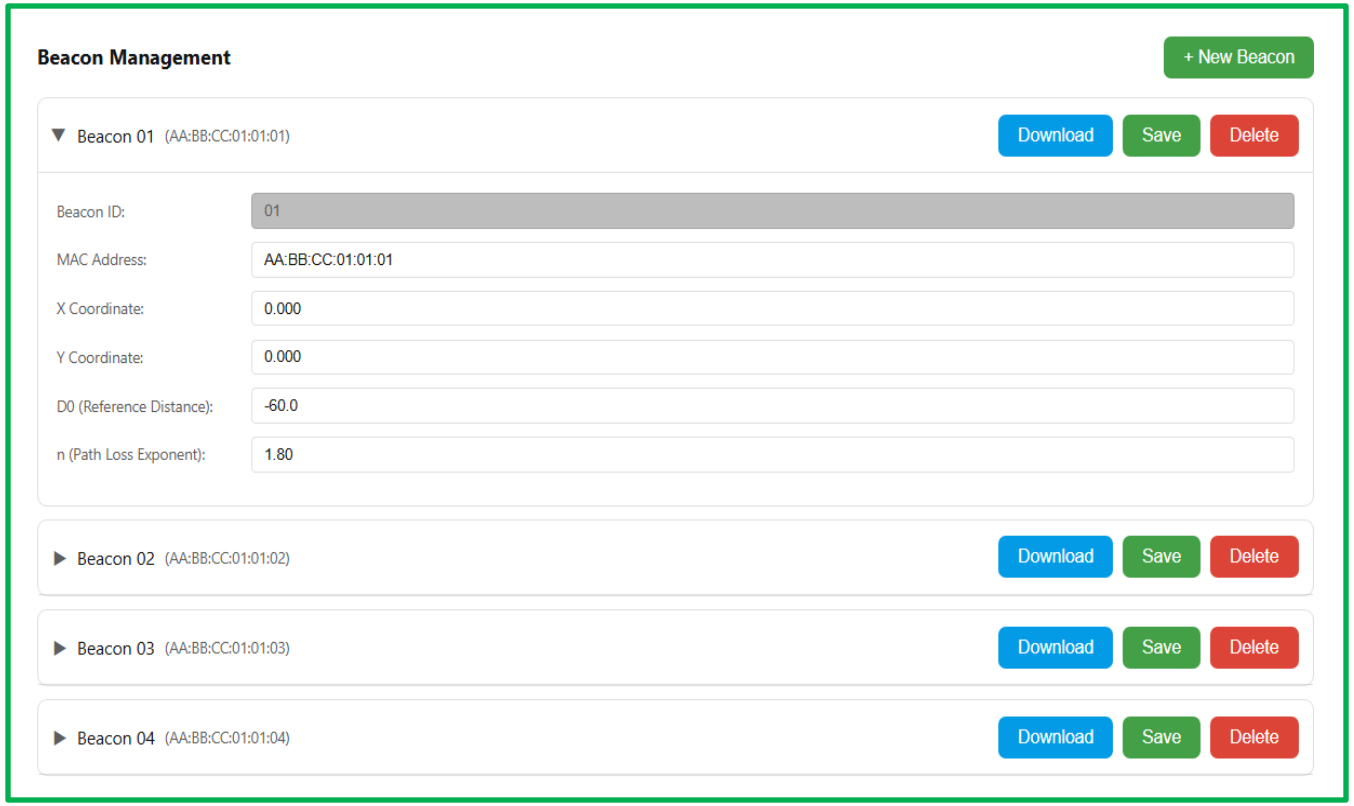
3. PRODUCT IMPLEMENTATION

3.2.3 Map manage-Admin Interface

Note: When adding a area, it must be a predefined from the map.



Define new area



Update beacon coordinates

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3. PRODUCT IMPLEMENTATION

3.3 Video Demo

https://drive.google.com/file/d/1lKnJ3hOE5-kbes9Hw6EU3SbCRiwT0VJ2/view?usp=drivesdk

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THANK
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